

Complete Electroweak/Yukawa LPM Matching for the q-D-H Portal

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral;
see RECOVERY_PROVENANCE.md.

Executive result

The simultaneous $SU(3)_c \times SU(2)_L \times U(1)_Y$ LPM calculation closes the v1.4 portal normalization band. Thermal masses are $m_H/T=0.438207$, $m_Q/T=0.503460$ and $m_D/T=0.418088$. The susceptibility-weighted Higgs occupation width is $\Gamma_H/T=3.603e-4$, giving $\Gamma_H/\Gamma_R=2.44e6$. Hard Yukawa-assisted $2<->2$ cuts remain a positive correction.

Reported benchmark values

- metric_1: 0.438207
- metric_2: 0.50346
- metric_3: 0.418088
- metric_4: 0.0003603
- metric_5: 36100.0
- metric_6: 2440000.0
- metric_7: 4.09e-07
- metric_8: 0.00529888708